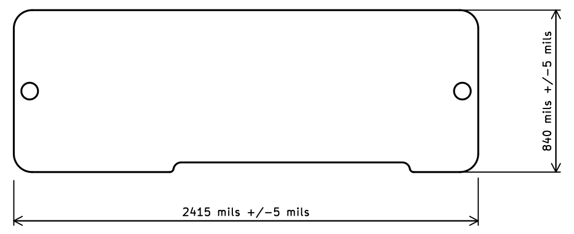
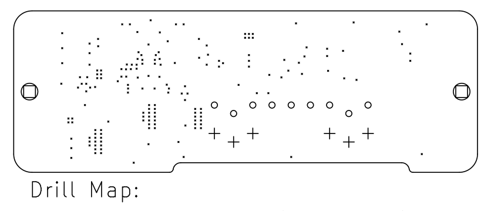
REVISION HISTORY				
REV	DESCRIPTION	DWN	CHK	APVD
1	ENGR RELEASE PER			ENG APPROVAL PER





Drill Map:

- · 0.152mm / 0.0060" (172 holes)
- ○ 0.813mm / 0.0320" (9 holes)
- + 1.448mm / 0.0570" (6 holes) (not plated)
- □ 2.200mm / 0.0866" (2 holes) (not plated)

HOLE TABLE

FINISHED DIAMETER	TOLERANCE	PLATED	CAPPED/ FILLED
0.006"	+0.003"/-0.006"	Y	Y
0.032"	+0.003"/-0.003"	Y	N
0.057"	+0.003"/-0.003"	N	N/A
0.0866"	+0.003"/-0.003"	N	N/A

LAYER STACKUP AND FILENAME

LAYER NAME	COPPER LAYER NUMBER	TYPE	FILENAME	THICKNESS (mils)	Er
FRONT SILKSCREEN		SILKSCREEN	100068610-001_Rev1-F_SILKSCREEN.GBR		
FRONT SOLDERMASK		SOLDERMASK	100068610-001_Rev1-F_MASK.GBR	1.00	4.50
FRONT COPPER	1	0.5 OZ COPPER	100068610-001_Rev1-F_CU.GBR	1.90	
370HR PP7628		PREPREG		8.10	4.26
370HR PP2116		PREPREG		4.80	4.14
INNER 1 COPPER	2	1 OZ COPPER	100068610-001_Rev1-IN1_CU.GBR	1.40	
370HR 4x7628		PREPREG		28.00	4.42
INNER 2 COPPER	3	1 OZ COPPER	100068610-001_Rev1-IN2_CU.GBR	1.40	
370HR PP2116		PREPREG		4.70	4.22
370HR PP7628		PREPREG		8.10	4.26
BACK COPPER	4	0.5 OZ COPPER	100068610-001_Rev1-B_CU.GBR	1.90	
BACK SOLDERMASK		SOLDERMASK	100068610-001_Rev1-B_MASK.GBR	1.00	4.50
BACK SILKSCREEN		SILKSCREEN	100068610-001_Rev1-B_SILKSCREEN.GBR		
			TOTAL THICKNESS	62.30 +/-5	
		BOARD OUTLINE	100068610-001_Rev1-EDGE_CUTS.GBR		
		PLATED THRU HOLES	100068610-001_Rev1-PTH.DRL		
		NON-PLATED THRU HOLES	100068610-001_Rev1-NPTH.DRL		

NOTE: FINISHED OUTER COPPER WEIGHT IS 1 OZ. THIS SHOULD BE 1/2 OZ BASE FOIL AND 1/2 OZ PLATING.

NOTES: UNLESS OTHERWISE SPECIFIED
1: STANDARDS
  1.1: THE PCB SHALL BE FABRICATED IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-6012 CLASS 2.
  1.2: INTERPRET DIMENSIONS AND TOLERANCES IN ACCORDANCE WITH THE CURRENT REVISION OF ASME Y14.5M.
  1.3: TEST IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-TM-650 2.4.22.
  1.4: DO NOT SCALE TO DRAWING.
2: MATERIAL
  2.1: THE PCB DIELECTRIC MATERIAL SHALL BE 370HR.
  2.2: THICKNESSES OF CONDUCTOR AND DIELECTRIC SHALL MATCH THE STACKUP DRAWING.
3: FLATNESS
  3.1: BOW AND TWIST OF ASSEMBLY SUB-PANEL OR SINGULATED PWB SHALL NOT EXCEED 0.75%.
  3.2: TEST IN ACCORDANCE WITH CURRENT REVISION OF IPC-TM-650 2.4.22.
4: ETCH GEOMETRY.
  4.1: MEASURE WIDTH FROM THE BASE OF THE METALLIZATION.
  4.2: FINISHED LINE WIDTH AND TERMINAL AREA SHALL NOT DEVIATE FROM THE GERBER PATTERN IMAGE BY MORE THAN 20%.
5: SURFACE FINISH
  5.1: FINAL FINISH SHALL BE IN ACCORDANCE WITH IPC-4552 AND SHALL BE ENIG.
6: HOLES
  6.1: SEE HOLE TABLE FOR DIMENSIONS (AFTER PLATING WHERE APPLICABLE).
  6.2: PLATING IN HOLES SHALL BE CONTINUOUS ELECTROLYTIC COPPER IN ACCORDANCE WITH IPC-6012, CLASS 2.
  6.3: SEE DRILL CHART FOR FINISHED HOLE SIZE AND PLATING.
  6.4: ALL HOLES SHALL BE LOCATED WITHIN 3 MIL OF TRUE POSITION AS SUPPLIED IN CAD DATA.
  6.5: THE TOLERANCE FOR ALL HOLES SHALL BE AS INDICATED IN THE HOLE TABLE.
  6.6: HOLES IN THE HOLE TABLE MARKED AS "CAPPED/FILLED" SHALL BE FILLED WITH A NON-CONDUCTIVE MATERIAL AND PLATED OVER.
7: SOLDERMASK
  7.1: SOLDERMASK OVER BARE COPPER (SMOBC) ON PRIMARY AND SECONDARY SIDES USING SUPPLIED ARTWORK IN ACCORDANCE WITH CURRENT REVISION OF IPC-SM-840 TYPE B.
  7.2: COLOR: GREEN.
  7.3: LIQUID PHOTO-IMAGEABLE (LPI) 0.001 INCH TO 0.002 INCH THICKNESS, HALOGEN FREE.
  7.4: NO BLEED-OUT ALLOWED OVER EXPOSED SMD PADS.
8: SILKSCREEN
  8.1: SILKSCREEN PRIMARY AND SECONDARY SIDE WITH NON-CONDUCTIVE, NON-NUTRIENT EPOXY INK.
  8.2: COLOR: WHITE.
  8.3: ANY UNSPECIFIED STROKE WIDTH SHALL BE 5 MIL.
  8.4: CLIP SILKSCREEN AWAY FROM ANY EXPOSED METAL.
9: FINISHING
  9.1: REMOVE ALL BURRS AND BREAK SHARP EDGES.
10: IMPEDANCE (ALL TOLERANCED +/- 10%)
  10.1: SEE IMPEDANCE REFERENCE TABLES.
  10.2: VENDOR MAY ADJUST DESIGN GEOMETRIES UP TO +/-20% TO ACHIEVE TARGET IMPEDANCE. ADJUSTMENTS BEYOND 20% OF LINE WIDTH, SPACING OR DIELECTRIC THICKNESS SHALL REQUIRE APPROVAL FROM ENGINEERING.
  10.3: VENDOR MAY SUBSTITUTE SPECIFIC CORES OR PREPREGS TO ACHIEVE IMPEDANCE TARGETS AS LONG AS THE DIELECTRIC REMAINS UNCHANGED.
11: PANELIZED PWBS LAYOUT
  11.1: IF THE PWB IS PANELIZED, THE VENDOR HAS THE FREEDOM TO MOVE INDIVIDUAL OR GROUPED PWBS TO ANY LOCATION ON THE PANEL TO EASE FABRICATION OR TEST.
12: NET TIES AND SHORTS
  12.1: CONSULT THE NET TIE SHORTS TABLE FOR NETS THAT ARE INTENTIONALLY SHORTED WITH NET TIES.
13: NON-DESTRUCTIVE EVALUATION
  13.1: ALL PWBS SHALL PASS 100% ELECTRICAL TEST USING SUPPLIED IPC-D-365 NETLIST IN ACCORDANCE WITH CURRENT REVISION OF IPC-9252, CLASS 2.
  13.2: CERTIFICATE OF CONFORMANCE SHALL BE SUPPLIED WITH EACH SHIPMENT.
  13.3: ALL BARE BOARDS THAT PASS ELECTRICAL TEST SHALL BE MARKED WITH WHITE EPOXY INK, IN ACCORDANCE WITH MIL-STD-130, WITH THE LETTERS 'ET'. MARKING SHALL NOT INTERFERE WITH ANY TRACE OR MOUNTING PAD.

NET TIE SHORTS

NETTIE NAME	X POS MILS	Y POS MILS	LAYER	NETS SHORTED
NT1	2237.00	1166.00	B.Cu	GNDA, GND
NT2	3563.00	1200.00	F.Cu	GNDD, GND

IMPEDANCE

TRACE LAYER	REF PLANE LAYER	IMPEDANCE (OHMS)	TRACE WIDTH (MILS)	TRACE SPACE (MILS)	TYPE
1	2	90	6.4	4.5	DIFF COPLANAR WAVEGUIDE W/GND

PART NO. 100068610-001

RELATED PARTS			
TYPE	NUMBER	CFG	REV
SCH	100046137	001	2
PWA	100068622	001	1
EBOM	100068622	001	1

DRAWING MEETS REQUIREMENTS OF MIL-STD-31000

DO NOT SCALE THIS DRAWING

UNLESS OTHERWISE SPECIFIED
CAPACITORS ARE IN MICROFARADS
RESISTORS ARE IN OHMS
DIMENSIONS ARE IN INCHES

TOLERANCE ON
DECIMALS ANGLES
± .01 ± .1°
± .005

CONTRACT NO.	
DWN	
CHK	
APVD	
APVD	
APVD	
APVD	
BY DIRECTION	DATE

PCB FABRICATION
SUPI PAYLOAD BOARD

SIZE	CAGE CODE	DRAWING NO.	REV
D		100068610	1

SCALE 1:1
FILENAME: 100068622-001_Rev1.kicad_pcb
SHEET: MANUFACTURING
SHEET: 1/1